NATURAL LANGUAGE PROCESSING

ASSIGNMENT 1

NLP TEXT PREPROCESSING

Submitted by,
Avinash V Bhaskaran
S7 CSE-AI
Roll No: 22

Aim

To perform text preprocessing on a self-introduction paragraph using Python and the NLTK library. The preprocessing steps include lowercase conversion, tokenization, punctuation removal, stop-word removal, stemming, and lemmatization.

1. Install NLTK

```
!pip install nltk
```

2. Import Required Libraries

```
import nltk
import string
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer
```

3. Download NLTK Resources

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('punkt_tab')
```

4. Self-Introduction Paragraph

```
paragraph = """
I am Avinash V Bhaskaran, a final-year student of Computer Science
and Engineering with a specialization in Artificial Intelligence. I
enjoy solving challenging problems and I am curious about how
```

```
machines learn from data.

During my studies I have built small projects in Python, including a
movie recommendation system and a chatbot for answering student
queries. These projects helped me understand data cleaning, model
training, and careful evaluation.

In my free time I like playing football, listening to music, and
reading about new research papers. I am especially interested in
Natural Language Processing and hands-on machine learning.

My strengths include curiosity, discipline, and clear communication
within a team. I always try to learn from my mistakes and share
knowledge with my friends.

One of my weaknesses is that I sometimes spend too much time on
small details, but I am learning to prioritize tasks better.
"""
```

5. Original Paragraph

```
print("Original Paragraph:\n")
print(paragraph)
```

```
Original Paragraph:

I am Avinash V Bhaskaran, a final-year student of Computer Science
and Engineering with a specialization in Artificial Intelligence. I
enjoy solving challenging problems and I am curious about how
machines learn from data.

During my studies I have built small projects in Python, including a
movie recommendation system and a chatbot for answering student
queries. These projects helped me understand data cleaning, model
training, and careful evaluation.

In my free time I like playing football, listening to music, and
reading about new research papers. I am especially interested in
Natural Language Processing and hands-on machine learning.

My strengths include curiosity, discipline, and clear communication
within a team. I always try to learn from my mistakes and share
knowledge with my friends.

One of my weaknesses is that I sometimes spend too much time on
small details, but I am learning to prioritize tasks better.
```

6. Lowercase Conversion

```
lower_text = paragraph.lower()
print(lower_text)
```

```
i am avinash v bhaskaran, a final-year student of computer science
and engineering with a specialization in artificial intelligence. i
enjoy solving challenging problems and i am curious about how
machines learn from data.

during my studies i have built small projects in python, including a
movie recommendation system and a chatbot for answering student
queries. these projects helped me understand data cleaning, model
training, and careful evaluation.

in my free time i like playing football, listening to music, and
reading about new research papers. i am especially interested in
natural language processing and hands-on machine learning.

my strengths include curiosity, discipline, and clear communication
within a team. i always try to learn from my mistakes and share
knowledge with my friends.

one of my weaknesses is that i sometimes spend too much time on
small details, but i am learning to prioritize tasks better.
```

7. Sentence Tokenization

```
sentences = sent_tokenize(lower_text)
print(sentences)
```

```
['i am avinash v bhaskaran, a final-year student of computer
science\nand engineering with a specialization in artificial
intelligence.', 'i\nenjoy solving challenging problems and i am
curious about how\nmachines learn from data.', 'during my studies i
have built small projects in python, including a\nmovie
recommendation system and a chatbot for answering
student\nqueries.', 'these projects helped me understand data
cleaning, model\ntraining, and careful evaluation.', 'in my free
time i like playing football, listening to music, and\nreading about
new research papers.', 'i am especially interested in\nnatural
language processing and hands-on machine learning.', 'my strengths
include curiosity, discipline, and clear communication\nwithin a
team.', 'i always try to learn from my mistakes and share\nknowledge
with my friends.', 'one of my weaknesses is that i sometimes spend
too much time on\nsmall details, but i am learning to prioritize
tasks better.']
```

8. Word Tokenization

```
words = word_tokenize(lower_text)
print(words)
```

```
['i', 'am', 'avinash', 'v', 'bhaskaran', ',', 'a', 'final-year',
'student', 'of', 'computer', 'science', 'and', 'engineering',
'with', 'a', 'specialization', 'in', 'artificial', 'intelligence',
'.', 'i', 'enjoy', 'solving', 'challenging', 'problems', 'and', 'i',
'am', 'curious', 'about', 'how', 'machines', 'learn', 'from',
'data', '.', 'during', 'my', 'studies', 'i', 'have', 'built',
'small', 'projects', 'in', 'python', ',', 'including', 'a', 'movie',
'recommendation', 'system', 'and', 'a', 'chatbot', 'for',
'answering', 'student', 'queries', '.', 'these', 'projects',
'helped', 'me', 'understand', 'data', 'cleaning', ',', 'model',
'training', ',', 'and', 'careful', 'evaluation', '.', 'in', 'my',
'free', 'time', 'i', 'like', 'playing', 'football', ',',
'listening', 'to', 'music', ',', 'and', 'reading', 'about', 'new',
'research', 'papers', '.', 'i', 'am', 'especially', 'interested',
'in', 'natural', 'language', 'processing', 'and', 'hands-on',
'machine', 'learning', '.', 'my', 'strengths', 'include',
'curiosity', ',', 'discipline', ',', 'and', 'clear',
'communication', 'within', 'a', 'team', '.', 'i', 'always', 'try',
'to', 'learn', 'from', 'my', 'mistakes', 'and', 'share',
'knowledge', 'with', 'my', 'friends', '.', 'one', 'of', 'my',
'weaknesses', 'is', 'that', 'i', 'sometimes', 'spend', 'too',
'much', 'time', 'on', 'small', 'details', ',', 'but', 'i', 'am',
'learning', 'to', 'prioritize', 'tasks', 'better', '.']
```

9. Punctuation Removal

```
words_without_punctuation = []
for word in words:
    if word not in string.punctuation:
        words_without_punctuation.append(word)
print(words_without_punctuation)
```

```
['i', 'am', 'avinash', 'v', 'bhaskaran', 'a', 'final-year',
'student', 'of', 'computer', 'science', 'and', 'engineering',
'with', 'a', 'specialization', 'in', 'artificial', 'intelligence',
'i', 'enjoy', 'solving', 'challenging', 'problems', 'and', 'i',
'am', 'curious', 'about', 'how', 'machines', 'learn', 'from',
'data', 'during', 'my', 'studies', 'i', 'have', 'built', 'small',
'projects', 'in', 'python', 'including', 'a', 'movie',
'recommendation', 'system', 'and', 'a', 'chatbot', 'for',
'answering', 'student', 'queries', 'these', 'projects', 'helped',
'me', 'understand', 'data', 'cleaning', 'model', 'training', 'and',
'careful', 'evaluation', 'in', 'my', 'free', 'time', 'i', 'like',
'playing', 'football', 'listening', 'to', 'music', 'and', 'reading',
'about', 'new', 'research', 'papers', 'i', 'am', 'especially',
```

```
'interested', 'in', 'natural', 'language', 'processing', 'and',
'hands-on', 'machine', 'learning', 'my', 'strengths', 'include',
'curiosity', 'discipline', 'and', 'clear', 'communication',
'within', 'a', 'team', 'i', 'always', 'try', 'to', 'learn', 'from',
'my', 'mistakes', 'and', 'share', 'knowledge', 'with', 'my',
'friends', 'one', 'of', 'my', 'weaknesses', 'is', 'that', 'i',
'sometimes', 'spend', 'too', 'much', 'time', 'on', 'small',
'details', 'but', 'i', 'am', 'learning', 'to', 'prioritize',
'tasks', 'better']
```

10. Stop Word Removal

```
stop_words = set(stopwords.words('english'))
filtered_words = []
for word in words_without_punctuation:
    if word not in stop_words:
        filtered_words.append(word)
print(filtered_words)
```

```
['avinash', 'v', 'bhaskaran', 'final-year', 'student', 'computer',
'science', 'engineering', 'specialization', 'artificial',
'intelligence', 'enjoy', 'solving', 'challenging', 'problems',
'curious', 'machines', 'learn', 'data', 'studies', 'built', 'small',
'projects', 'python', 'including', 'movie', 'recommendation',
'system', 'chatbot', 'answering', 'student', 'queries', 'projects',
'helped', 'understand', 'data', 'cleaning', 'model', 'training',
'careful', 'evaluation', 'free', 'time', 'like', 'playing',
'football', 'listening', 'music', 'reading', 'new', 'research',
'papers', 'especially', 'interested', 'natural', 'language',
'processing', 'hands-on', 'machine', 'learning', 'strengths',
'include', 'curiosity', 'discipline', 'clear', 'communication',
'within', 'team', 'always', 'try', 'learn', 'mistakes', 'share',
'knowledge', 'friends', 'one', 'weaknesses', 'sometimes', 'spend',
'much', 'time', 'small', 'details', 'learning', 'prioritize',
'tasks', 'better']
```

11. Stemming

```
stemmer = PorterStemmer()
print("Stemming Results:\n")
for word in filtered_words:
    print(word, "->", stemmer.stem(word))
```

```
Stemming Results:

avinash -> avinash
v -> v
bhaskaran -> bhaskaran
```

```
final-year -> final-year
student -> student
computer -> comput
science -> scienc
engineering -> engin
specialization -> special
artificial -> artifici
intelligence -> intellig
enjoy -> enjoy
solving -> solv
challenging -> challeng
problems -> problem
curious -> curiou
machines -> machin
learn -> learn
data -> data
studies -> studi
built -> built
small -> small
projects -> project
python -> python
including -> includ
movie -> movi
recommendation -> recommend
system -> system
chatbot -> chatbot
answering -> answer
student -> student
queries -> queri
projects -> project
helped -> help
understand -> understand
data -> data
cleaning -> clean
model -> model
training -> train
careful -> care
evaluation -> evalu
free -> free
time -> time
like -> like
playing -> play
football -> footbal
listening -> listen
music -> music
reading -> read
new -> new
research -> research
papers -> paper
especially -> especi
interested -> interest
natural -> natur
```

```
language -> languag
processing -> process
hands-on -> hands-on
machine -> machin
learning -> learn
strengths -> strength
include -> includ
curiosity -> curios
discipline -> disciplin
clear -> clear
communication -> commun
within -> within
team -> team
always -> alway
try -> tri
learn -> learn
mistakes -> mistak
share -> share
knowledge -> knowledg
friends -> friend
one -> on
weaknesses -> weak
sometimes -> sometim
spend -> spend
much -> much
time -> time
small -> small
details -> detail
learning -> learn
prioritize -> priorit
tasks -> task
better -> better
```

12. Lemmatization

```
lemmatizer = WordNetLemmatizer()
print("Lemmatization Results:\n")
for word in filtered_words:
    print(word, "->", lemmatizer.lemmatize(word))
```

```
Lemmatization Results:

avinash -> avinash
v -> v
bhaskaran -> bhaskaran
final-year -> final-year
student -> student
computer -> computer
science -> science
engineering -> engineering
```

```
specialization -> specialization
artificial -> artificial
intelligence -> intelligence
enjoy -> enjoy
solving -> solving
challenging -> challenging
problems -> problem
curious -> curious
machines -> machine
learn -> learn
data -> data
studies -> study
built -> built
small -> small
projects -> project
python -> python
including -> including
movie -> movie
recommendation -> recommendation
system -> system
chatbot -> chatbot
answering -> answering
student -> student
queries -> query
projects -> project
helped -> helped
understand -> understand
data -> data
cleaning -> cleaning
model -> model
training -> training
careful -> careful
evaluation -> evaluation
free -> free
time -> time
like -> like
playing -> playing
football -> football
listening -> listening
music -> music
reading -> reading
new -> new
research -> research
papers -> paper
especially -> especially
interested -> interested
natural -> natural
language -> language
processing -> processing
hands-on -> hands-on
machine -> machine
learning -> learning
```

```
strengths -> strength
include -> include
curiosity -> curiosity
discipline -> discipline
clear -> clear
communication -> communication
within -> within
team -> team
always -> always
try -> try
learn -> learn
mistakes -> mistake
share -> share
knowledge -> knowledge
friends -> friend
one -> one
weaknesses -> weakness
sometimes -> sometimes
spend -> spend
much -> much
time -> time
small -> small
details -> detail
learning -> learning
prioritize -> prioritize
tasks -> task
better -> better
```

13. Final Preprocessed Text

```
lemmatized_words = []
for word in filtered_words:
    lemmatized_words.append(lemmatizer.lemmatize(word))
print("Final Preprocessed Text:\n")
for word in lemmatized_words:
    print(word, end=" ")
```

```
Final Preprocessed Text:

avinash v bhaskaran final-year student computer science engineering
specialization artificial intelligence enjoy solving challenging
problem curious machine learn data study built small project python
including movie recommendation system chatbot answering student
query project helped understand data cleaning model training careful
evaluation free time like playing football listening music reading
new research paper especially interested natural language processing
hands-on machine learning strength include curiosity discipline
clear communication within team always try learn mistake share
knowledge friend one weakness sometimes spend much time small detail
learning prioritize task better
```

GitHub Repository

https://github.com/AVINASH-V-BHASKARAN/NLP_Text_Processing

Conclusion

The self-introduction paragraph was successfully preprocessed using Python and the NLTK library. The preprocessing steps included lowercase conversion, sentence and word tokenization, punctuation removal, stop-word removal, stemming, and lemmatization. The final output produced a clean and refined version of the text suitable for NLP applications.